
Classic iPod Battery Door Mechanism Design Proposal

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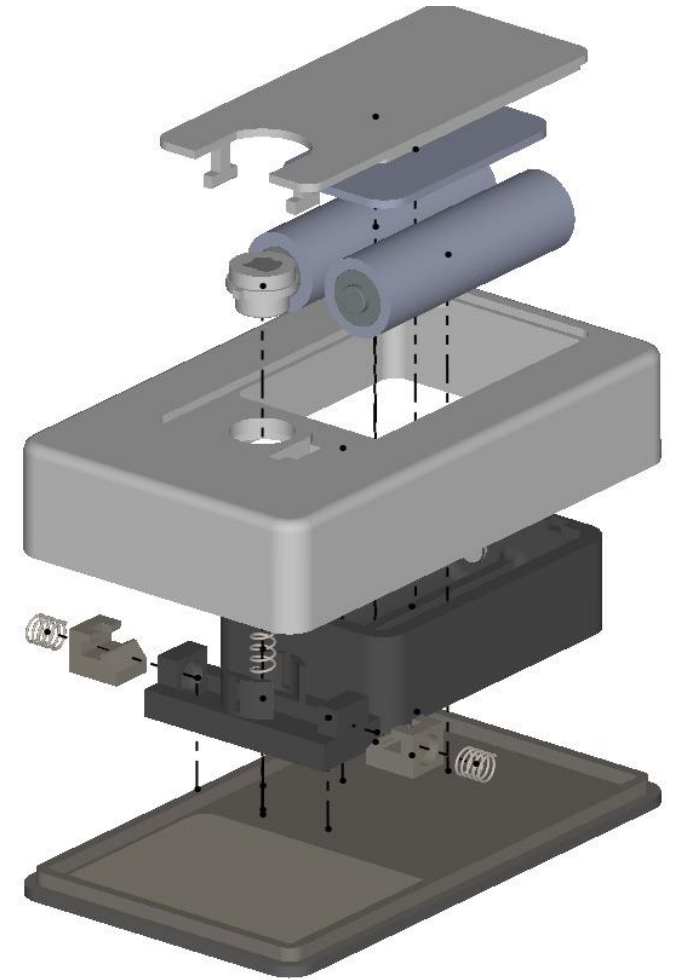
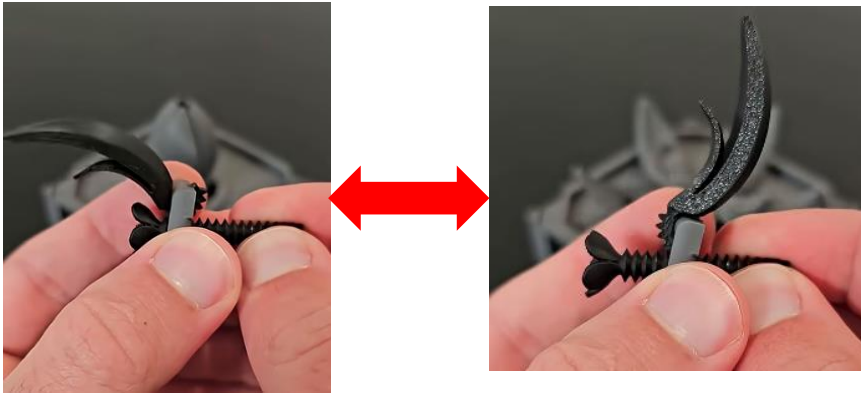


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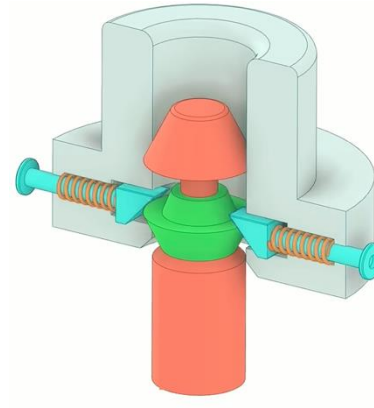
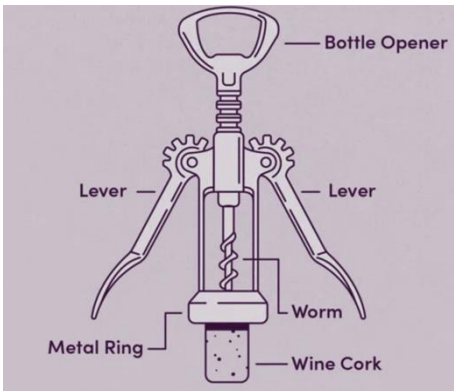
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Inspiration

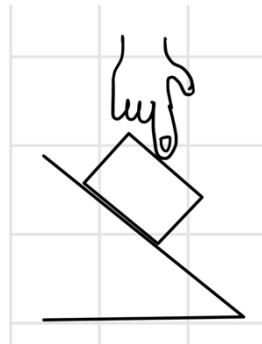
Actuation



Rack and Pinion

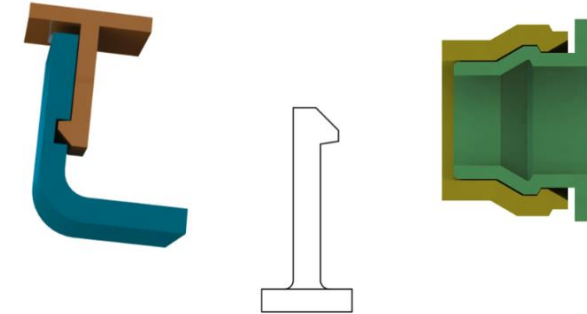


Auto Lock



Force on an Inclined surface

Securing



Snap-fits / cantilevers



Sliding pins



Magnets

Images from: [Rack and pinion flower](#), [Auto Lock](#), [Chain door guard](#), [Wine opener](#), [Snap fit](#), [magnets](#)

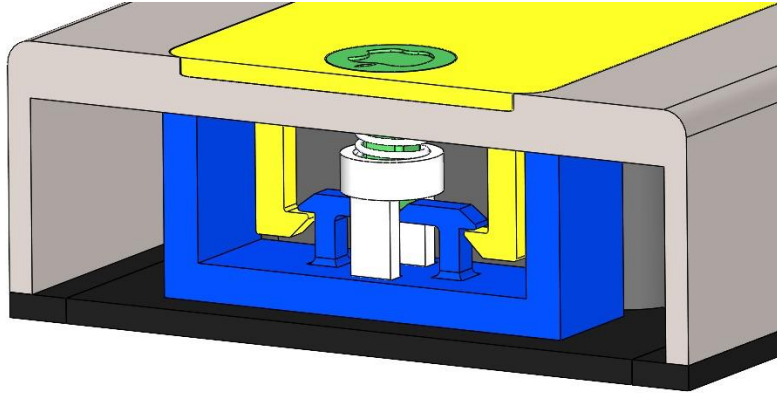
Mechanism Selection

Snap fit		Magnet	
Pro	Con	Pro	Con
- Simple mechanism and design - Low cost and easy manufacturability - Fast assembly with no additional hardware	- Relies on elastic deformation - Creep and fatigue from repetitive use may cause failure - Can break during separation	- Simple design with clean aesthetic - Good resistance to normal separation	- Requires magnetic field containment (design becomes less compact) - Magnet strength degradation from temperature differential - Adds material costs and potential interference with nearby electronics
Rack and pinion		Inclined Surfaces	
Pro	Con	Pro	Con
- Precise and repeatable - Controlled displacement	- Wear over time - Requires tight tolerances and alignment - May require lubrication	- Simple mechanism and design - Compact	- Friction affects actuation force consistency - Sensitive to surface finish

Note:

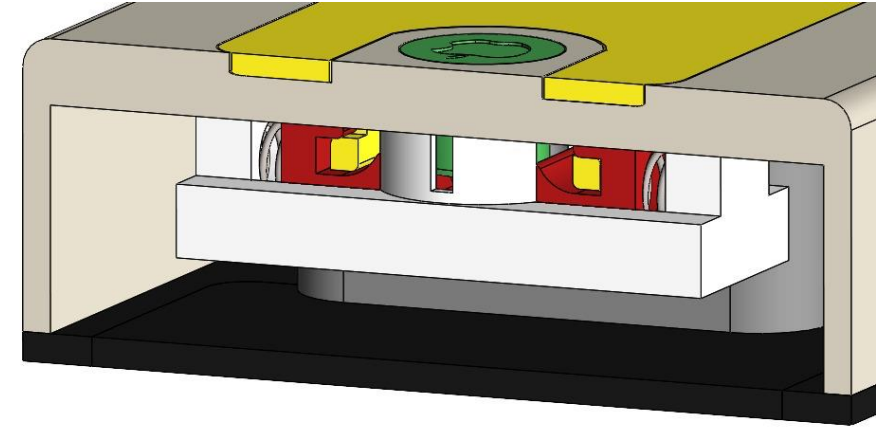
All mechanisms reach the same functional outcome for the battery door. It is important to prioritize longevity, simplicity, aesthetic elegance, and impact on other components

Mechanism Selection



Elastic Bending Arms

Pros	Cons
Minimal parts	Relies on elastic deformation
Simple geometry	Creep and fatigue
Fast assembly	Actuation force varies with material and aging
Tactile feedback during actuation	Higher risk of fracture during misuse or forced separation



Inclined Surface

Pros	Cons
Controlled and repeatable actuation	Slightly increased part complexity
Durable and long cycle life	Requires tighter control of interface geometry
Consistent performance across temperature and aging	Sensitive to surface finish and friction

- Choose Design 2 
- Added part complexity is minimal and remains compatible with injection molding
- Friction effects at inclined interfaces can be effectively managed through surface finish and material selection
- Controlled and consistent performance and durability over the product's lifetime.
- Note: Components' colors have been altered to enhance visibility and referenceability. They do not reflect the intended color post-manufacturing

Design 1

Bending beams

h : vertical button travel

$F_{applied}$: user actuation force

k_{button} : button spring stiffness

δ : flexure-arm deflection

δ_{min} : minimum flexure-arm deflection required for disengagement

k_{arm} : stiffness of one flexure arm

k_{flex} : equivalent stiffness of both arms together

L : Cantilever arm length

b : Cantilever arm width

t : Cantilever arm thickness

$$W_{applied} = W_{button\ spring} + W_{flex}$$

$$W_{applied} = F_{applied} * h$$

$$W_{flex} = U_{flex} = \int_0^L \frac{M(x)^2}{2EI} dx \approx \frac{1}{2} * k_{flex} * \delta^2$$

$$\therefore F_{applied} * h = \frac{1}{2} * k_{button} * h^2 + \frac{1}{2} * k_{flex} * \delta^2$$

$$\rightarrow F_{applied} = \frac{1}{2} * k_{button} * h + \frac{1}{2} * k_{flex} * \frac{\delta^2}{h}$$

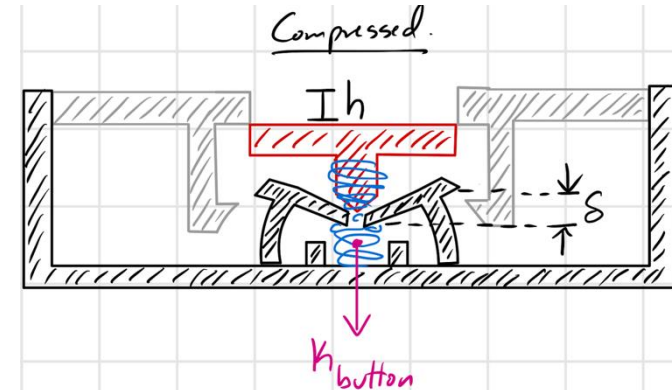
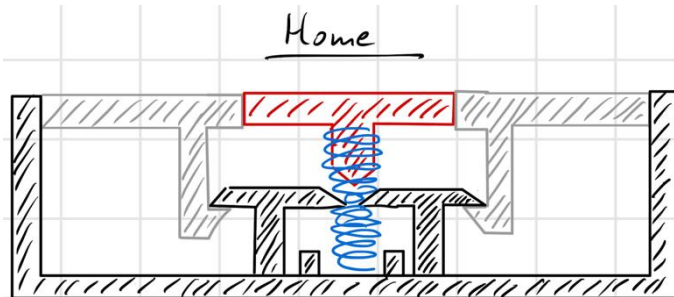
$$\text{Where, } k_{flex} = 2 * k_{arm} = 2 * \frac{3EI}{L^3} \quad \text{and} \quad I = \frac{b*t^3}{12}$$

Assumptions

- Negligible friction and energy loss
- Input work goes into spring energy and flexure energy
- Quasi-static actuation—inertia negligible
- Button spring only compresses in vertically
- Linear spring behavior
- Small deflection beam bending
- Beam bending energy modeled as spring
- Symmetric flexure arms share load equally

Disengagement occurs when flexure deflection reaches the required clearance $\delta \geq \delta_{min}$

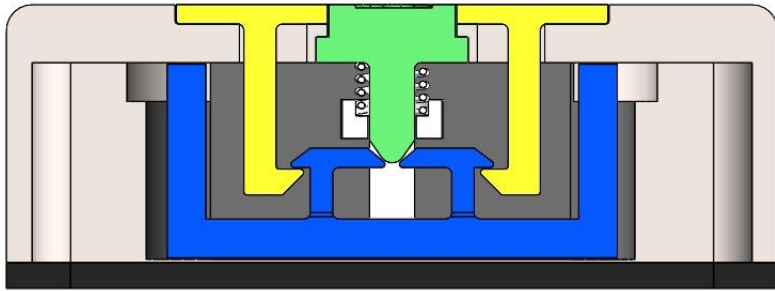
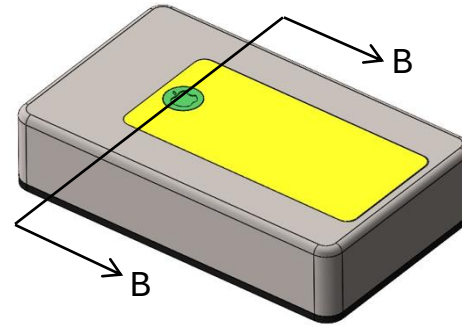
Optimize variable k_{button} , k_{flex} , and δ based on design requirements



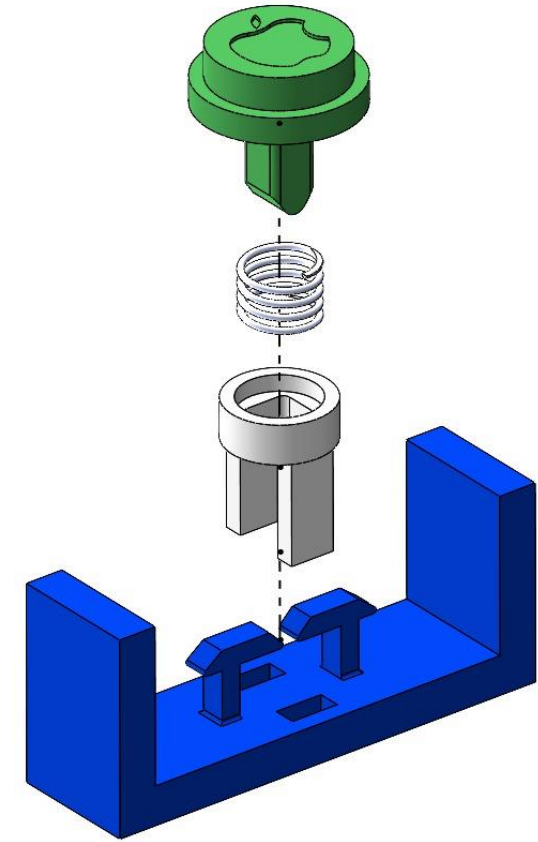
Design 1: Exploration

Legend

Beige:	Back Casing
Yellow:	Battery Door
Grey:	Battery Housing
Green:	Button
White:	Button Frame
Blue:	Bending Arms
Black:	iPod Front



Section B-B



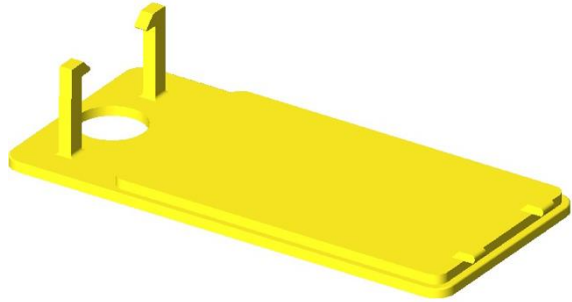
Mechanism Assembly

How it works:

- User applies a downward force on the button, and the load is transferred to the integrated bending arms
- Bending arms elastically deflect downward, disengaging the door retention features and releasing the battery door
- Once released, the door can be lifted out of its seated position by the user
- Upon reinsertion, the bending arms elastically deflect again and snap back into place, re-capturing the door and restoring the locked state

Design 1: Important Components

Battery Door



Material & Manufacturing

- Injection molded PC
- Hard coat finish

Unit Price

- PC: ~\$5.5 per kg
- Mass of part: 7.34 g
- Volume: 6165.81 mm^3
- Hard coat finish (assumption): \$0.5
- Unit price: \$0.54

Button



Material & Manufacturing

- Injection molded PC
- Hard coat finish

Unit Price

- PC: ~\$5.5 per kg
- Mass of part: 0.6 g
- Volume: 501.89 mm^3
- Hard coat finish (assumption): \$0.5
- Unit price: \$0.5

Button Frame



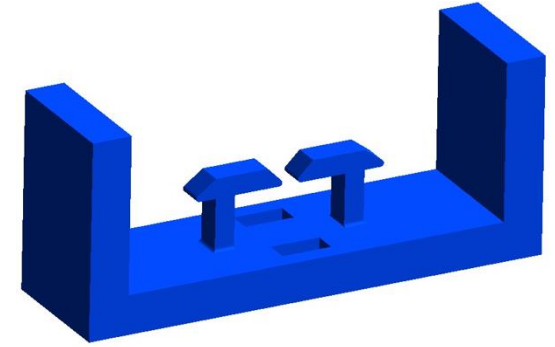
Material & Manufacturing

- Injection molded ABS

Unit Price

- ABS: \$1.77 per kg
- Mass of part: 0.21 g
- Volume: 173.32 mm^3
- Unit price: \$0.001 (rounded for clarity)

Bending Arm



Material & Manufacturing

- Injection molded ABS
- Will require side actions for bending arms

Unit Price

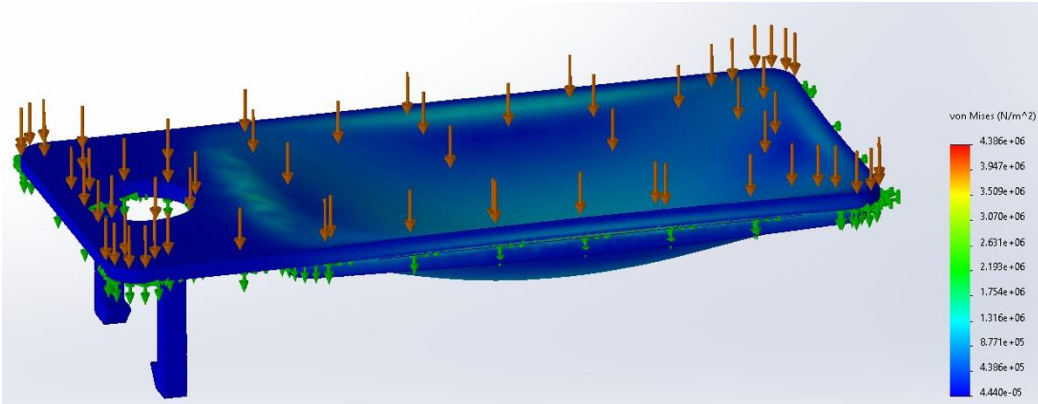
- ABS: \$1.77 per kg
- Mass of part: 2.15 g
- Volume: 1808.21 mm^3
- Unit price: \$0.004

Note:

- Cost analysis does not include quotes of injection molding
- CAD of iPod chassis (screen and back casing) excluded
- CAD of Battery Casing excluded—shared with design 2 and expanded on later

Design 1: FEA

Battery Door



70 N force normal to surface (Extreme case)

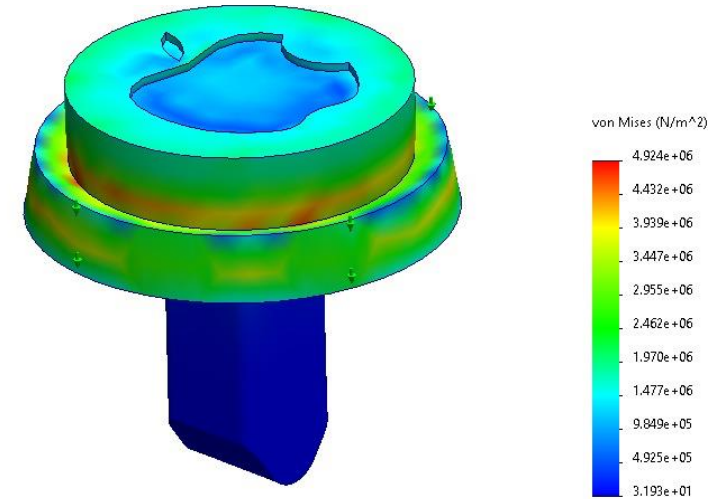
Max: 4.38 MPa (von Mises)

PC Yield Strength: [55, 75] MPa

$$\text{Safety Factor (SF)} = \frac{60\text{MPa}}{4.38\text{MPa}} \approx 13.7$$

Design is safe ✓

Button



70 N force normal to button surface (Extreme case)

Max: 4.92 MPa (von Mises)

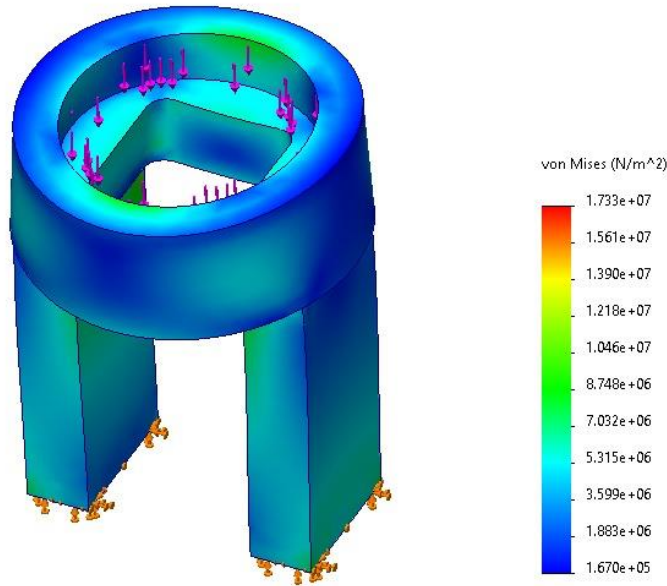
PC Yield Strength: [55, 75] MPa

$$SF = \frac{60\text{MPa}}{4.92\text{MPa}} \approx 12.2$$

Design is safe ✓

Design 1: FEA

Button Frame

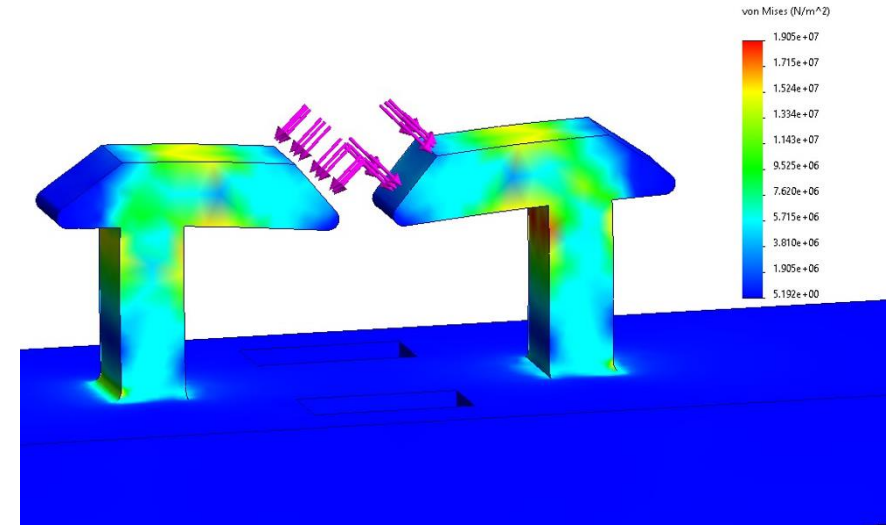


50 N force normal to contact surface (Extreme Case) Max: 17.33 MPa (von Mises)
ABS Yield Strength: [29.6, 48] MPa

$$SF = \frac{40 \text{ MPa}}{17.33 \text{ MPa}} \approx 2.3$$

Design is safe ✓

Bending Arm



7 N force normal to contact surface (~10N equivalent vertical force) Max: 19.05 MPa (von Mises)
ABS Yield Strength: [29.6, 48] MPa

$$SF = \frac{40 \text{ MPa}}{19.05 \text{ MPa}} \approx 2.1$$

Design is safe ✓

Design 2

Force on an Inclined Surface

h : vertical button travel

$F_{applied}$: user actuation force

k_{button} : button spring stiffness

k_{clasp} : clasp spring stiffness

x : clasp horizontal travel

θ : wedge/ramp angle

$$\tan\theta = \frac{h}{x} \leftrightarrow h = x * \tan\theta$$

$$W_{applied} = W_{button} + 2W_{clasp}$$

$$W_{applied} = F_{applied} * h$$

$$W_{button} = \frac{1}{2} * k_{button} * h^2$$

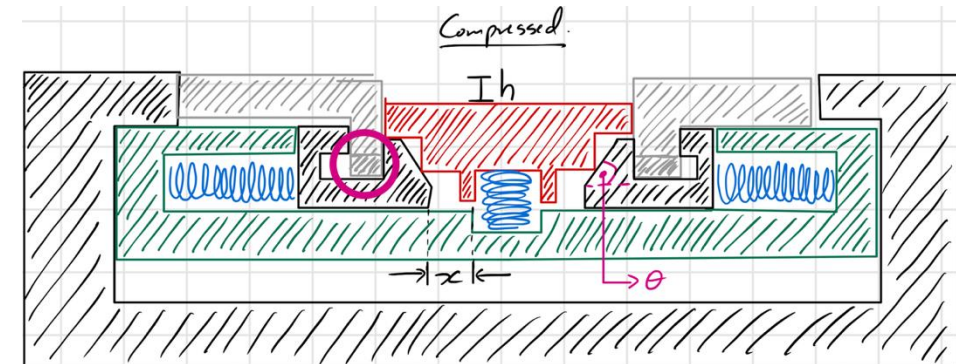
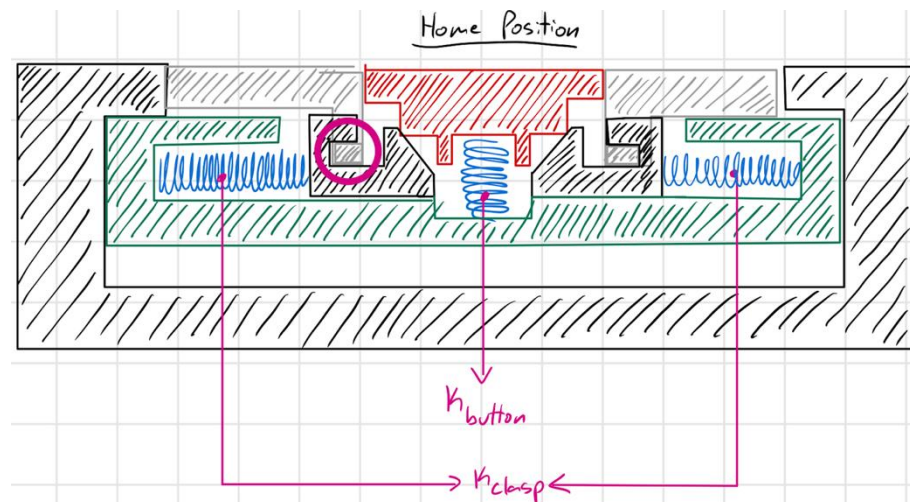
$$W_{clasp} = \frac{1}{2} * k_{clasp} * x^2$$

$$F_{applied} * h = \frac{1}{2} * k_{button} * h^2 + 2 * \left(\frac{1}{2} * k_{clasp} * x^2 \right)$$

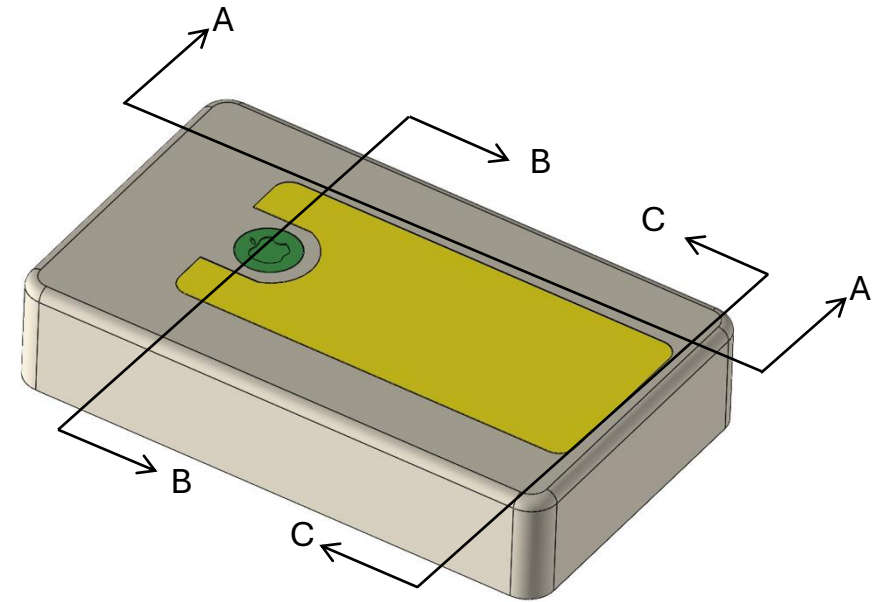
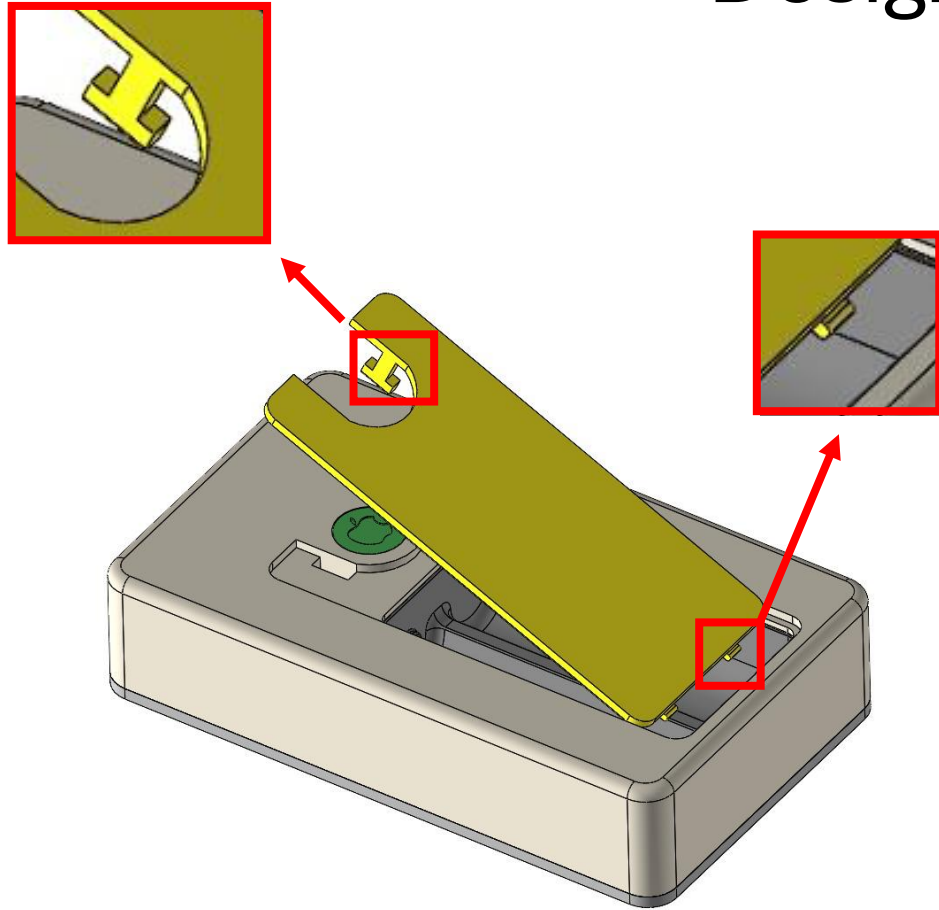
$$\rightarrow F_{applied} = \frac{1}{2} * k_{button} * x * \tan\theta + \frac{k_{clasp} * x}{\tan\theta}$$

Assumptions

- Negligible friction
- All input work goes into spring energy
- 2 clasp springs have symmetric loading and same displacement
- Quasi-static actuation—inertia negligible
- Button spring only compresses vertically
- Clasp springs only compress laterally
- Linear spring behavior



Design 2: Exploration

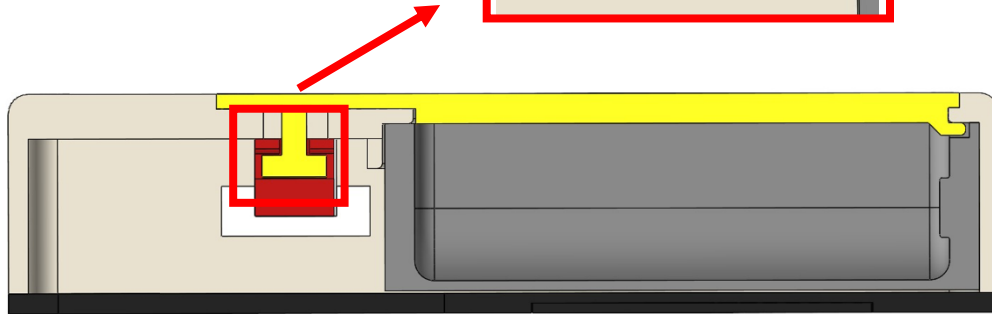
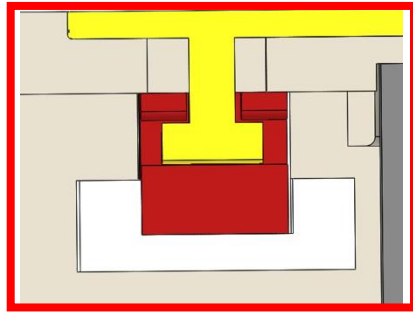


The Battery Door is held in place by two pivot tabs and two extrusions

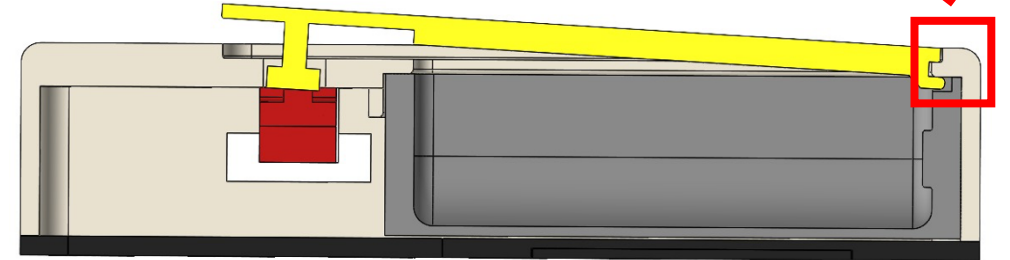
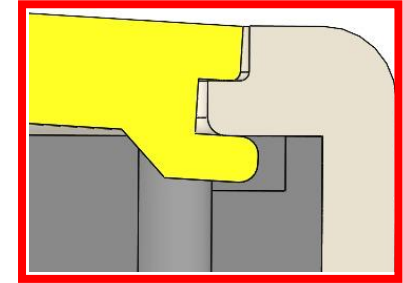
Design 2: Exploration

Legend

Beige:	Back Casing
Yellow:	Battery Door
Grey:	Battery Housing
Green:	Button
White:	Button frame
Red:	Clasp
Black:	iPod Front



Section A-A, Battery Door Secured



Section A-A, Battery Door Detached

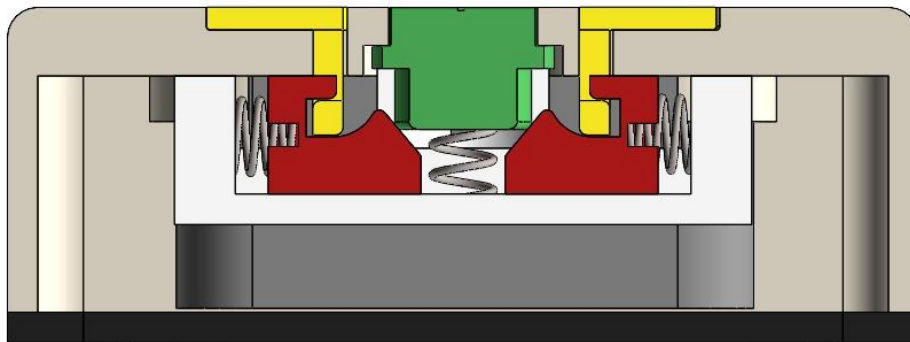
Notes

- Pivot tabs have fillets to ensure seamless insertion and removal. Can be deburred further if required
- Pivot tabs of the Battery Door are held securely between Battery Housing and the Back Casing (close-up on the right)
- The Battery Door has extrusions that slide into and are held in place by the Clasp (close-up on the left)
- Batteries and Padding are **excluded** to enhance visibility. Assume polyurethane Padding stuck onto the inside surface of the Battery Door
- Cavity between Back Casing, Button Frame, and iPod Front is assumed to be filled with electronic components (not shown)
- Assume coincident surface mate between Button Frame and Back Casing using a thin layer of B7000 or T7000 adhesive (exact surface shown in following slides)

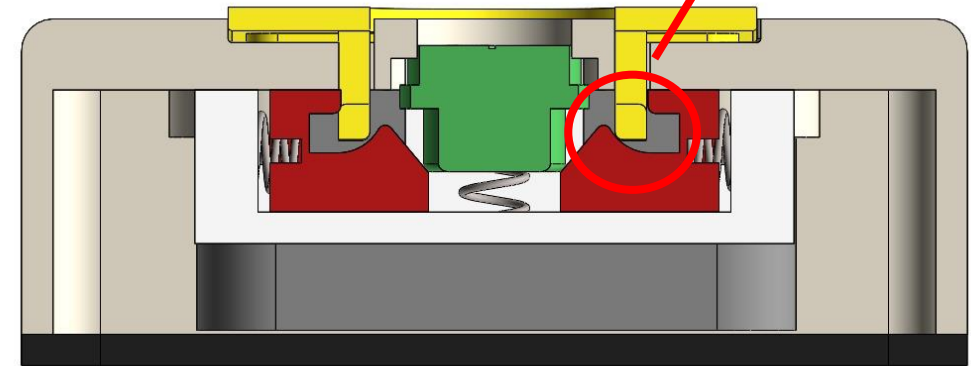
Design 2: Exploration

Legend

Beige:	Back Casing
Yellow:	Battery Door
Grey:	Battery Housing
Green:	Button
White:	Button frame
Red:	Clasp
Black:	iPod Front



Section B-B, Battery Door Secured



Section B-B, Battery Door Detached

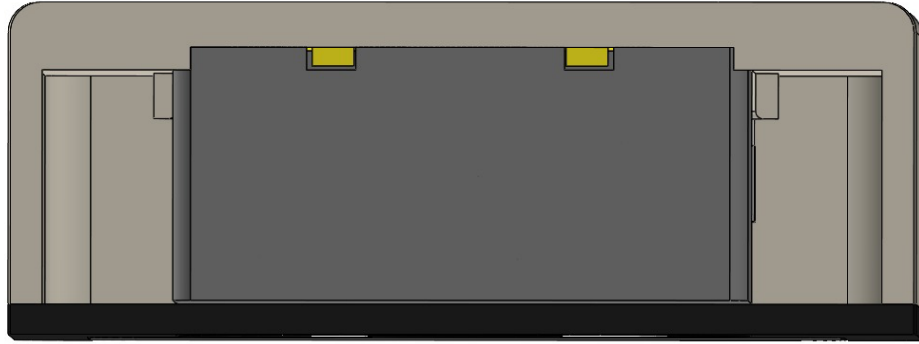
How it works:

- A downward Button press drives the clasps laterally outward
- The filleted Clasp surfaces engage the Battery Door extrusions, converting lateral motion into a controlled upward lift
- This motion releases the Battery Door and provides sufficient clearance for user removal

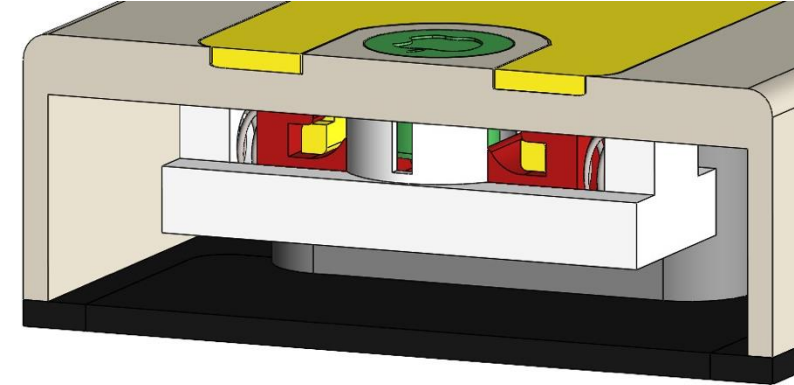
Design 2: Exploration

Legend

Beige:	Back Casing
Yellow:	Battery Door
Black:	iPod Front



Section C-C, Battery Door Secured

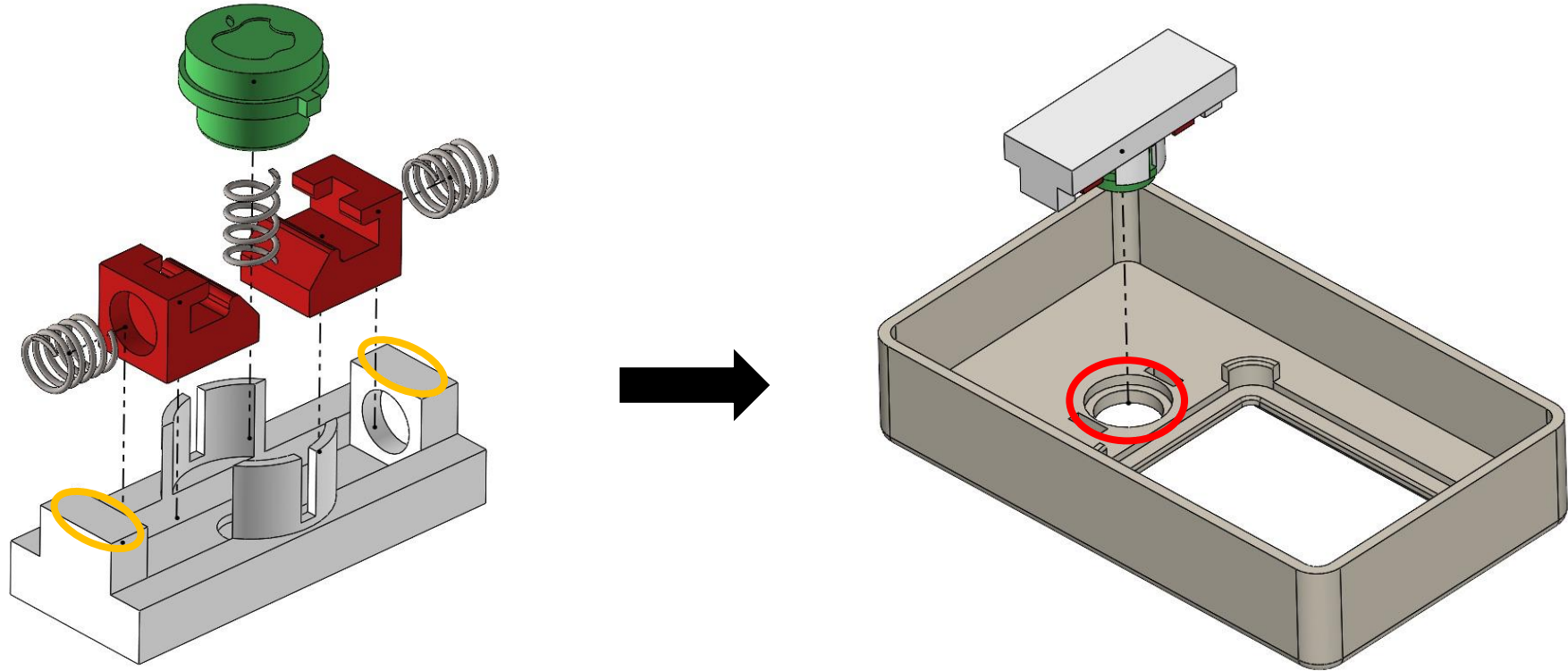


Notes

- Pivot tabs of the Battery Door rest in a cavity created between the Battery Housing and the Back Casing
- The tabs define the door's rotation axis and prevent vertical translation during normal use
- ~0.25mm of clearance is provided to accommodate for manufacturing tolerances

Legend	
Beige:	Back Casing
Green:	Button
White:	Button frame
Red:	Clasp

Design 2: Assembling the Mechanism



To prevent the button from being detached from the assembly by the spring forces, a circular ledge in the Back Casing—**highlighted by the red circle**—was added

Assumption:

The surface mate between the Back Casing and the Button Frame is assumed to be held in place once electronic components are added. For extra stability, a thin layer of B7000 or T7000 adhesive can be applied on the mating surface—**highlighted by the orange circles**—of the Button Frame

Reference Values
 $h = 1.8\text{mm}$
 $F_{\text{applied}} = 7\text{N}$

Design 2: Button Design

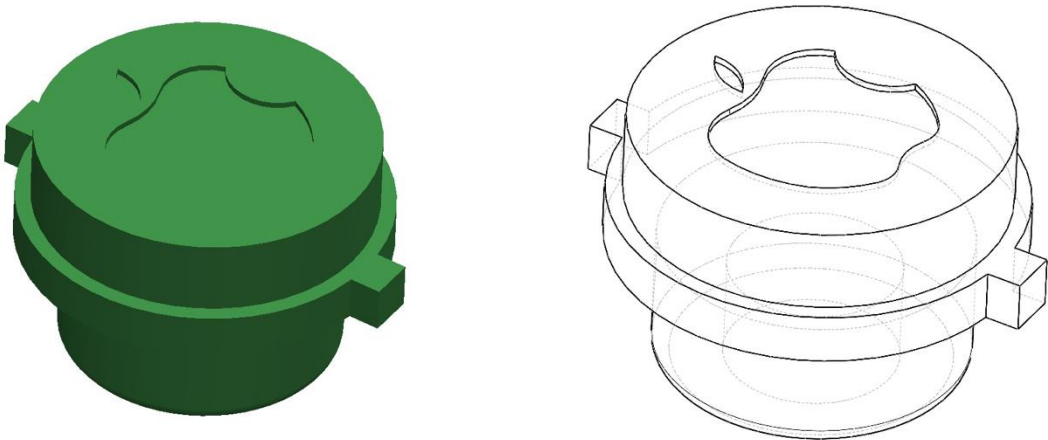
	DIMENSIONS (Diameter, D)						RESISTANCE		
	Fingertip		Thumb		Palm		Single Finger	Different fingers ¹	Thumb/ Palm
	Bare hand	Gloved hand	Bare hand	Gloved hand	Bare hand	Gloved hand			
MIN	10 mm (0.4")	19 mm (0.75")	19 mm (0.75")	25 mm (1.0")	40 mm (1.6")	50 mm (2.0")	2.8 N (10 oz)	1.4 N (5 oz)	2.8 N (10 oz)
MAX	25 mm (1.0")	—	25 mm (1.0")	—	70 mm (2.8")	—	11.0 N (40 oz)	5.6 N (20 oz)	23.0 N (80 oz)

	DISPLACEMENT (A)	
	Fingertip	Thumb or Palm
MIN	2 mm (0.08")	3 mm (0.12")
MAX	6 mm (0.25")	38 mm (1.5")

	SEPARATION (S)				
	Single Finger		Single Finger Sequential	Different Fingers	Thumb or Palm
	Bare	Gloved			
MIN	13 mm (0.5")	25 mm (1.0")	6 mm (0.25")	6 mm (0.25")	25 mm (1.0")
PREF	50 mm (2.0")	—	13 mm (0.5")	13 mm (0.5")	150 mm (6.0")

¹Actuated at same time
NOTE: Where gloved hand criteria are not provided, minima should be suitably adjusted.

FIGURE 12. Push button (finger or hand operated)
[DoD Design Criteria Standard](#)



Button Vertical Displacement

- Buttons are commonly designed with vertical displacement of [2, 6] mm.
- Measurements of the buttons on Apple products show that the actual mechanical travel is $\leq 1\text{ mm}$.
- A **vertical displacement of 1.8 mm** was chosen to balance guidelines with mainstream design.

Button Force

- Guidelines indicates actuation force range of [2.8, 11.0] N.
- 7.0 N, middle value of this range, used for calculations

Button Size

- Guidelines indicate fingertip diameter from [10, 25] mm
- Button size chosen as **11 mm** ensuring comfortable and reliable operation

Button Material & Manufacturing Selection

- **Injection molded PC with hard coat finish**
- High impact resistance and stiffness. Hard coating improves scratch resistance thus maintaining aesthetics

Button Costs

- PC: ~\$5.5 per kg
- Hard coat finish (assumption): \$0.5
- Volume: 605.88 mm^3
- Mass: 0.72 g
- Unit cost: \$0.5

Note: cost analysis does not include quotes of injection molding

Reference Values

$$h = 1.8\text{mm}$$

$$F_{\text{applied}} = 7\text{N}$$

$$\theta = 50^\circ$$

$$x = 1.55\text{mm}$$

$$k_{\text{button}} \approx 1.22\text{ kN/m}$$

$$k_{\text{clasp}} \approx 1.8\text{ kN/m}$$

Design 2: Clasp & Spring Design

$$F_{\text{applied}} = \frac{1}{2} * k_{\text{button}} * x * \tan\theta + \frac{k_{\text{clasp}} * x}{\tan\theta}$$

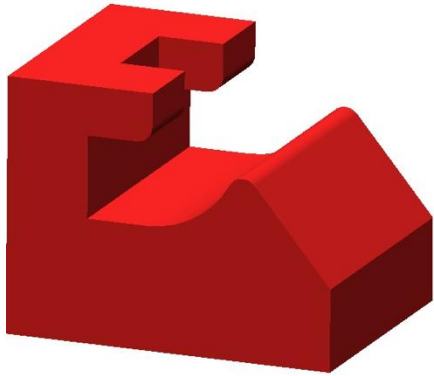
$$h = x * \tan\theta$$

Clasp wedge angle, θ , horizontal displacement, x , of clasp, and spring constants

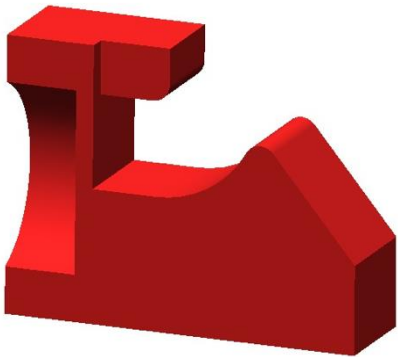
- Use reference values for F_{applied} and h , and use calculus to find optimal θ

$$\frac{dF_{\text{applied}}}{d\theta} = x \sec^2\theta \left(\frac{1}{2} k_{\text{button}} - \frac{k_{\text{clasp}}}{\tan^2\theta} \right)$$

- For balanced spring stiffness, $\theta \approx [45, 50]^\circ$ reduces sensitivity of actuation force to angular tolerances and limit excessive lateral loading
- Choose 50° as injection molded parts do not hit nominal geometry exactly
- Resulting, $x = 1.55\text{mm}$
- Minimum-norm / optimized solution: $k_{\text{button}} \approx 1.22\text{ kN/m}$ and $k_{\text{clasp}} \approx 1.80\text{ kN/m}$



Full View



Cross-Sectional View

Clasp Material & Manufacturing Selection

- Injection molded PC

Clasp Costs

- PC: ~\$5.5 per kg
- Volume: 363.19 mm^3
- Mass: 0.43 g
- Unit cost: \$0.0024

Note: cost analysis does not include quotes of injection molding

Spring Costs

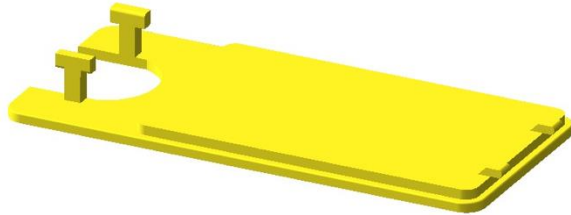
- ~\$0.10 per spring

Note:

- Average cost assumed from [Alibaba](#)
- More costs incurred for custom springs

Design 2: Other Components

Battery Door



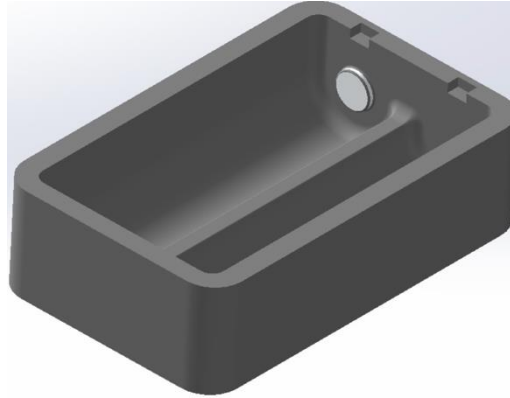
Material & Manufacturing

- Injection molded PC
- Hard coat finish

Unit Price

- PC: ~\$5.5 per kg
- Mass of part: 7.21 g
- Volume: 6057.11 mm^3
- Hard coat finish (assumption): \$0.5
- Unit price: \$0.54

Battery Casing



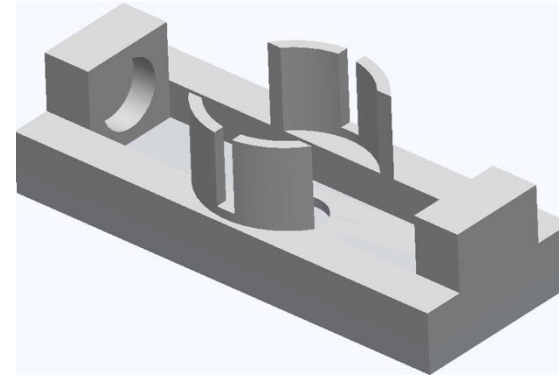
Material & Manufacturing

- Injection molded ABS

Unit Price

- ABS: \$1.77 per kg
- Mass of part: 14.87 g
- Volume: 14581.35 mm^3
- Unit price: \$0.026

Button Frame



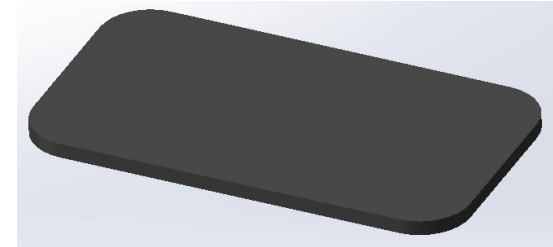
Material & Manufacturing

- Injection molded ABS
- Will require side actions for transverse holes

Unit Price

- ABS: \$1.77 per kg
- Mass of part: 2.5g
- Volume: 2448.25 mm^3
- Unit price: \$0.004

Padding



Material & Manufacturing

- Die cut polyurethane

Unit Price

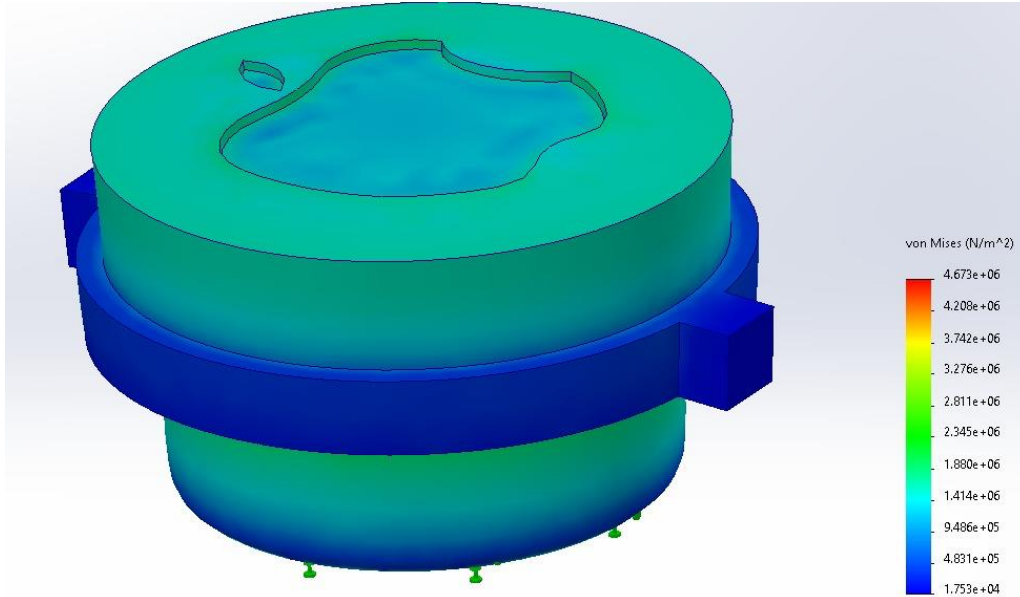
- Polyurethane: ~\$3 per kg
- Mass of part: 2.03g
- Volume: 1655.31 mm^3
- Die cut (assumption): \$0.01
- Unit price: \$0.016

Note:

- Cost analysis does not include quotes of injection molding
- CAD of iPod chassis (screen and back casing) excluded

Design 2: FEA

Button



100 N force normal to button surface (Extreme case)

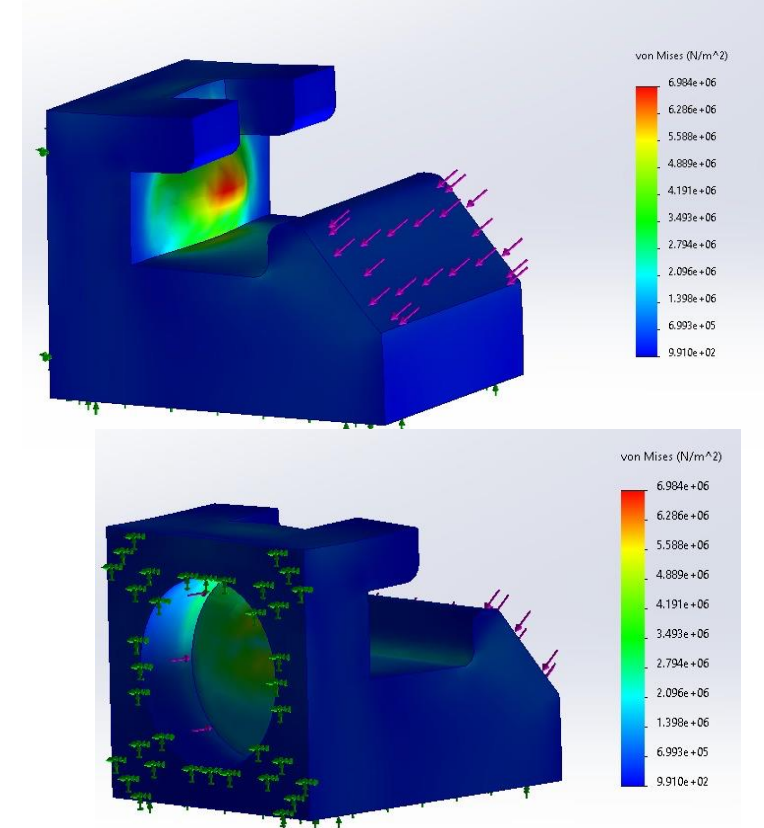
Max: 4.67 MPa (von Mises)

PC Yield Strength: [55, 75] MPa

$$\text{Safety Factor (SF)} = \frac{60 \text{ MPa}}{4.67 \text{ MPa}} \approx 12.8$$

Design is safe ✓

Clasp



20 N force normal to contact surface (Extreme case)

20N force to replicate spring (Extreme case)

Max: 6.98 MPa (von Mises)

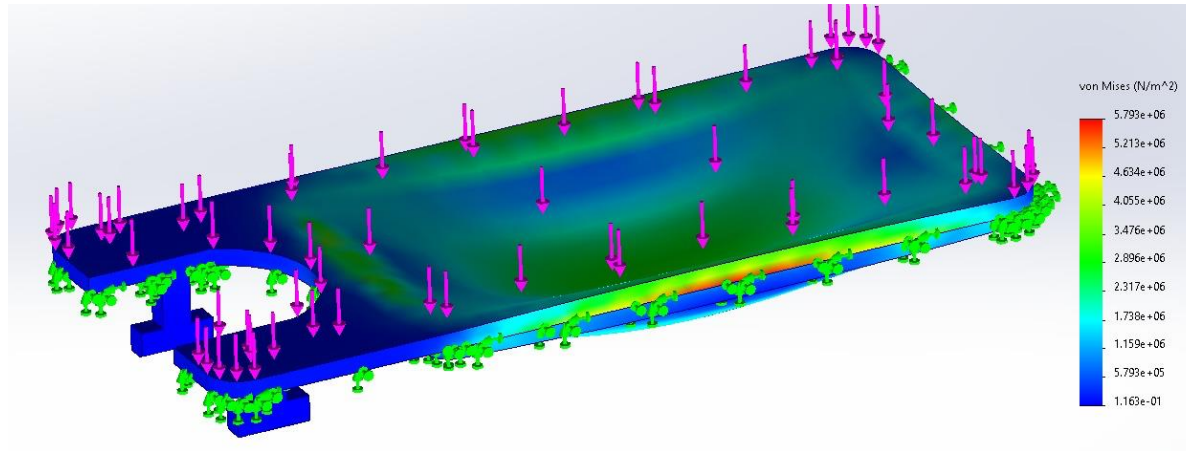
ABS Yield Strength: [29.6, 48] MPa

$$\text{SF} = \frac{40 \text{ MPa}}{6.98 \text{ MPa}} \approx 5.7$$

Design is safe ✓

Design 2: FEA

Battery Door



200 N force normal to battery door surface(Extreme case)

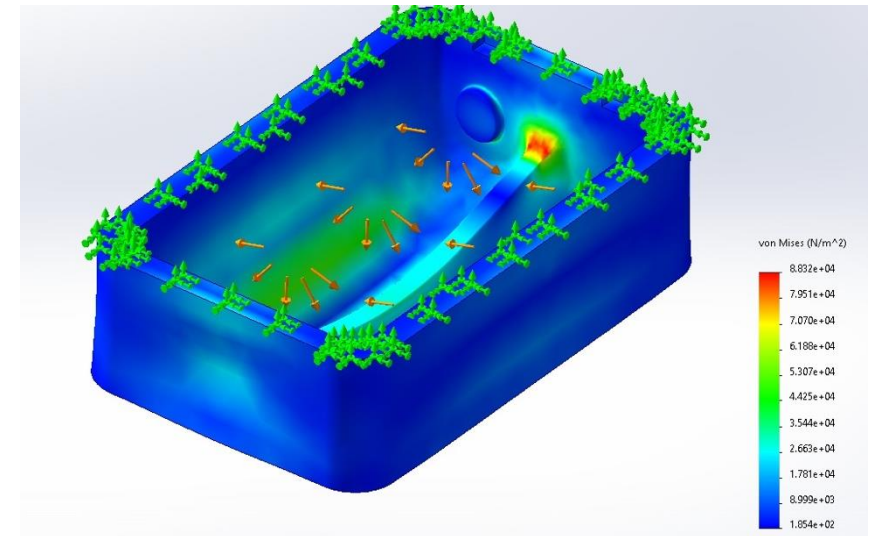
Max: 5.8 MPa (von Mises)

PC Yield Strength: [55, 75] MPa

$$SF = \frac{60MPa}{5.8MPa} \approx 10$$

Design is safe ✓

Battery Casing



50 N force radially in battery cavity (Extreme case)

Max: 2.21 MPa (von Mises)

ABS Yield Strength: [29.6, 48] MPa

$$SF = \frac{40MPa}{2.21MPa} \approx 18$$

Design is safe ✓

Bill of Materials

Bill of Materials for Design 2				
Number	Part Name	Quantity	Material	Unit Price (\$)
1	Battery Casing	1	ABS	0.026
2	Battery Door	1	Hard Coated PC	0.54
3	Button	1	Hard Coated PC	0.5
4	Button Frame	1	ABS	0.004
5	Clasp	2	PC	0.0024
6	Padding	45mm x 25mm x 1.5mm	Polyurethane	0.016
7	Spring (button)	1	Stainless Steel	0.1
8	Spring (clasp)	2	Stainless Steel	0.1

Note: BoM does not include iPod front / back casing, and batteries—assume that chassis and vendor components are in inventory and incur no extra cost